

		Signal Source				
		State ____	IMU Gyro	IMU Af*	IMU Ar*	Encoder
Controlled Value	Travel Distance	(integral of Vf)		Technically		Yes (l/r average, using accumulation)
	Forward Velocity	Yes		Technically		Yes (l/r average)
	Forward Accel	(derivative of Vf)		Yes		Technically
	Heading	Yes	(integral of Gz)			Yes (l/r delta, using accumulation)
	Yaw Rate	Yes	Yes		If paired with a Vf indicator	Yes (l/r delta)
	Alpha	(derivative of Yaw Rate)	(derivative of Gz)			Technically

*: compensated/corrected values required, as the accelerometer reads a mix of acceleration, yaw rate, forward velocity, and alpha